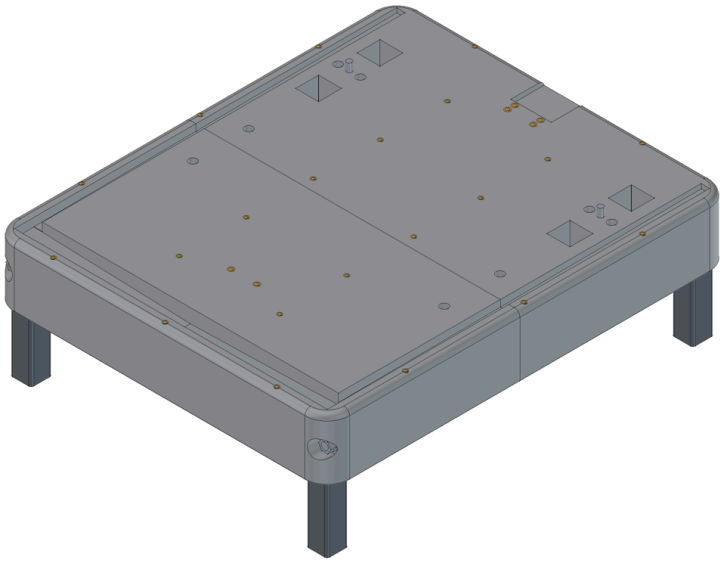
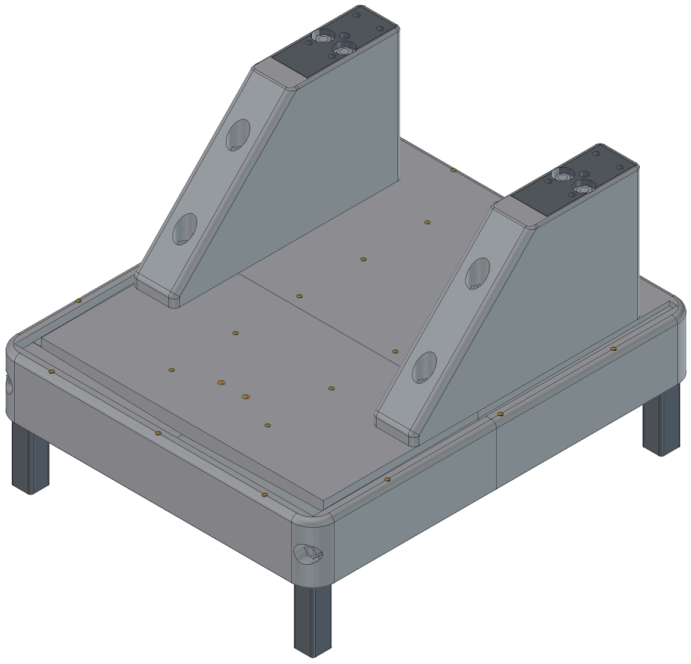
Build guide 3: frame and rail risers

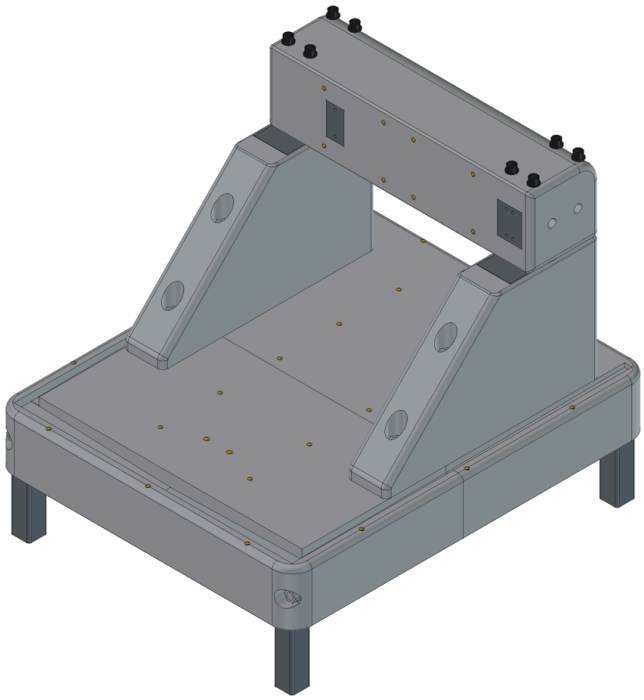
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4. Assemble the frame

1. **Feet:** set the rear base half on the bench on its legs. Bond a 1/2-13 nut into a printed plug in each leg end (or use the threaded inserts), screw in the leveling feet, and roughly level it.
2. **Join the base halves:** glue the O-ring cord into the rear half's split groove (glue the cord ends). Push the front half on (its tubes and sleeves over the spigots) and tighten the two 1/2-13 draw rods from the front pockets (snug, then about 40 N·m). Level the whole base with the feet.
3. **Grout the posts.** The sockets are 1.5 mm a side bigger than the posts, so the fit comes from epoxy, not from the two molds agreeing. Wax the posts (three coats, buffed) and the base dowels. Mix a little of the leftover epoxy with whiting to a thick cream and half-fill each socket and each upright's dowel hole. Glue the foot O-ring into each upright's groove, lower the upright on (posts into the sockets, foot on the base), thread the four tie rods into the coupling nuts and snug them, wipe off what squeezes out, and leave it 24 h. After that the posts lift out and drop back in with no play.
4. **Uprights:** add washers and nuts in the pockets (two at the top, two in the sloped front) and tighten to about 40-50 N·m, working front to back. Push the printed caps into the two sloped pockets, flush with the face (a dab of silicone holds them).
5. **Beam:** the top plates' dowel holes are 14 mm, oversize the same way. Wax the beam's dowels, put grout in the holes, set the beam on and fit the eight 3/8-16 x 6 in bolts snug. Check the beam face is square to the base top with a square and a dial indicator (shim under one top plate if needed) and leave it 24 h before the final torque (about 40 N·m).







5. Bed the rail risers

The risers are bonded to the castings with a thin epoxy grout and held down by screws into the cast-in inserts. Nothing is screwed into the cast-in tubes; the tubes stiffen the casting and form the joints.

1. Abrade and degrease the casting under each riser and the riser's underside.
2. Set the riser on small shim stacks (feeler stock) at each hold-down. Fit the M5 low-head screws (M5 x 25 for Y, M5 x 20 for X) just snug.
3. **Y risers (base):** make each one straight along its length and both coplanar with each other, within 0.02 mm over the length if you can. Check with the straightedge and feelers, or a dial indicator on a bridge spanning both. Set them parallel, 180 mm apart on center (rail spacing), using the rails as a gauge.
4. **X risers (beam face):** same, 100 mm apart. Make them parallel to the base top, checked with the indicator on the Y-axis table later, or now with a height gauge from the base top.
5. Dam the ends and sides with tape or plasticine, and flow low-viscosity epoxy thickened with a little whiting underneath from one side until it comes out the other.
6. After 48 h, torque the hold-downs to about 4 N·m (brass inserts) and recheck flatness.

